International Journal of Human–Computer Interaction

ISSN: 1044-7318 (Print) 1532-7590 (Online) Journal homepage: www.tandfonline.com/journals/hihc20

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To cite this article: Yu Zhang, Yang Bian, Xiaohua Zhao, Xuewei Li & Jianhua Zhang (2025) Improving Pedestrian Safety with Head-Up Display Warning in a Connected Environment, International Journal of Human–Computer Interaction, 41:6, 3724-3744, DOI: 10.1080/10447318.2024.2368910

To link to this article: https://doi.org/10.1080/10447318.2024.2368910

Published online: 17 Sep 2024.

INTERNATIONAL JOURNAL OF HUMAN–COMPUTER INTERACTION
2025, VOL. 41, NO. 6, 3724–3744
https://doi.org/10.1080/10447318.2024.2368910

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Improving Pedestrian Safety with Head-Up Display Warning in a Connected Environment

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ABSTRACT

In this paper, the potential of using a head-up display (HUD) in the connected environment to improve a vehicle's running comfort and pedestrian safety is tested, by providing warning information to drivers in advance. To achieve this objective, driving simulation technology is used to construct the connected environment and develop the HUD, and the effectiveness of the system is then tested. Specifically, thirty-four participants were recruited to conduct driving simulation experiments in six scenarios: three warning display types (Baseline/Head-down display/Head-up display) combined with two weather conditions (clear weather/foggy weather). The effects of the three different warning display types on braking risk-avoidance strategy were studied by comparing the drivers' performance during the perception and decision stage (position of accelerator-pedal release, position of first braking), the risk-avoidance manipulation stage (maximum deceleration, braking distance) and the risk-avoidance result stage (minimum collision distance, position of minimum speed). The influences of weather conditions and driver attributes were also considered. When the HUD warnings were activated, drivers started to decelerate further away from pedestrians, with a more stable and moderate deceleration process and a greater safety margin between the vehicle and the pedestrians. Using HUD warnings in foggy conditions improved drivers' perception and decision abilities, this study confirmed the great benefits that HUD warnings in the connected environment can bring to traffic safety, especially under risky situations and inclement weathers. The research results provide a reference for the more humanized and rationalized optimization design of these warning systems.

KEYWORDS

Connected environment; head-up display warning; risk avoidance strategy; pedestrian safety; generalized linear mixed model

1. Introduction

A connected environment collects information related to surrounding traffic by means of vehicle-to-vehicle(V2V), vehicle-to-infrastructure(V2I), and vehicle-to-everything (V2X) communications (Ali et al., 2020; Maag et al., 2022). Such information, in the form of in-vehicle warnings, is expected to improve drivers' situational awareness of surrounding traffic during driving, such as potentially dangerous vehicles and pedestrians (Aust et al., 2013; Cicchino, 2017). As a traditional human-computer interaction interface, head-down display (HDD) is installed on the dashboard or control panel of vehicles, requiring drivers to lower their heads to obtain information (Liu & Wen, 2004). In contrast, head-up display (HUD) projects the required information directly into the drivers' line of vision, reducing eye adjustment time and the possibility of ignoring dangerous stimuli on the road (Ablaßmeier et al., 2007; Liu, 2003). It is reasonable to expect that HUD warning system in the connected environment has great potential to improve driving behavior.

In recent years, pedestrian safety has received extensive attention due to its high accident rate and mortality. A report in the United States mentioned that it is estimated that 6,590 pedestrians died in motor vehicle accidents in 2019, based on the preliminary data on pedestrian traffic deaths in all fifty states and the District of Columbia (Retting & GHSA, 2020). In pedestrian and motor vehicle accidents, pedestrian crossing is the main source of risk. Pedestrian crossing refers to the behavior of pedestrians crossing roads (Wu et al., 2013), where pedestrian crossing locations that may conflict with motor vehicles typically include zebra crossings at intersections, zebra crossings on roads, and roads outside the zebra crossing. Among them, pedestrian crossing events outside the zebra crossing pose a higher risk and are also one of the most common emergencies. According to statistical data from Italy in 2020 (ACI-ISTAT, 2017), 7,204 (3.2%) road accidents were caused by pedestrians' inappropriate behavior when crossing the road outside the crosswalk. Therefore, pedestrian crossing events outside the zebra crossing are one of the most important challenges facing road safety and the focus of this article.

The HUD warning system in the connected environment seems to have brought new possibilities to address pedestrian safety issues. Limited research has shown that compared to no warning or HDD warning, the HUD warning increases the likelihood that drivers would glance toward latent the pedestrian (Hajiseyedjavadi et al., 2018), shortens

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braking response time (Winkler et al., 2018b), and increases safety margins with pedestrians (Naujoks & Neukum, 2014). By reviewing previous research, it has been found that the influence of HUD warning systems on driving behavior has been preliminarily explored, while the impact of HUD warning systems on driver risk avoidance strategies still needs further exploration. Risk avoidance strategies can reflect the driving behavior of drivers in various stages such as perception, judgment, decision, and manipulation during the risk avoidance process, as well as the final risk avoidance results (Yue et al., 2020). The optimal risk avoidance strategy can improve the stability of the risk avoidance process while ensuring safety. Therefore, before the large-scale application of HUD, it is necessary to fully explore how the HUD warning system affects drivers' risk avoidance strategies and whether it can help drivers choose the optimal risk avoidance strategy.

In fact, the effectiveness of the HUD warning system is also influenced by complex external conditions and driver attributes. Regarding the complex external environment, it includes weather conditions (e.g., foggy, rain and snow), linear road conditions (e.g., sharp curves and steep slopes) and light conditions (day and night). As a typical inclement weather with reduced visibility, foggy weather diminished the drivers' visual distance and perception of the surrounding environment, which had a significant negative impact on driver behavior (Gao et al., 2020; Mueller & Trick, 2012). Previous studies have shown that HUD warning system in the connected environment can effectively avoid vehicle collisions at intersections and rear-end collisions at sections under foggy conditions, helping drivers respond earlier and brake smoother (Wu et al., 2018; Zhang et al., 2020). It is worth further exploring whether the HUD pedestrian warning system is equally effective in foggy weather. In addition, there are differences in the degree to which drivers with different attributes (e.g., gender, age and driving experience) benefit from the HUD warning system. Previous research has suggested that HUD warning systems may be more effective for inexperienced drivers (Li et al., 2020). When using the HUD driving assistance system, male and female drivers exhibit different horizontal and longitudinal driving behaviors (Lyu et al., 2019). Therefore, when exploring the impact of the HUD warning system on driver braking risk avoidance strategies, it is meaningful to consider its interaction with weather conditions or driver attributes.

This study aims to explore the comprehensive effectiveness of HUD warning systems in avoiding pedestrian collisions in the connected environment. Specifically, the effect of the HUD warning on drivers' braking risk-avoidance strategy in the pedestrian crossing event outside the zebra crossing in the connected environment was explored through the driver recognition process in spatial dimension, considering the interaction of weather and driver attributes. To achieve this objective, an advanced driving simulator was used to build the connected environment and to test the effectiveness of the HUD warning on the pedestrian crossing event outside the zebra crossing under six scenarios of three warning display types (Baseline/HDD/HUD) $\times$ two weather conditions (clear weather/foggy weather). A comprehensive index system was constructed from the spatial dimension to reflect the entire process of braking risk-avoidance. Finally, the generalized linear mixed model was applied to preliminarily explore the effects of warning display types, weather conditions, driver attributes (gender, driving experience, driving frequency) and their interaction factors on the braking risk-avoidance strategy, and deeply analyze the differences in the influence of different levels of the same factor on the braking risk-avoidance strategy. The proposed method can provide theoretical supports for the research of braking risk-avoidance strategy, and the results may provide the basis for the optimization and design of HUD warning systems in the connected environment.

2. Literature review

This section is divided into four subsections: 2.1 The overall development and application of HUD; 2.2 Application of HUD warning in avoiding pedestrian collisions; 2.3 Test tool for HUD warning utility; 2.4 A method of exploring multiple influencing factors.

2.1. The overall development and application of HUD

HUD was first applied in the aviation industry to assist pilots. Nowadays, HUD has been widely used in-vehicle information systems. It uses the principle of optical reflection to project driving-related information to the front of the driver's line of sight, reducing the frequency of the driver's head lowering, improving information reading efficiency, improving driving distraction and load, and effectively ensuring driving safety. As an in-vehicle information system, HUD was initially used to display navigation information, vehicle status information, and entertainment communication information. With the development of connected technology and the realization of V2X communication, HUD has been widely used to display safety assistance information, supporting various safety-critical scenarios such as vehicle collisions, bicycle collisions, obstacle collisions and pedestrian collisions (Winkler et al., 2018b; Z. Yang et al., 2019). However, when HUD provides information that is unreasonable or unclear, it may lead to driver distraction and misunderstanding, resulting in negative effects (Ahmed et al., 2020; Pfannmüller et al., 2015; Z. Yang et al., 2019). Therefore, developing a reasonable HUD warning system and testing its effectiveness in various safety-critical scenarios have become a focus of scholars' current attention.

2.2. Application of HUD warning in avoiding pedestrian collisions

Pedestrians are considered one of the most vulnerable road users, with characteristics such as being difficult to observe and difficult to predict trajectories, leading to high accident and mortality rates. HUD warning systems have great potential in improving pedestrian safety through V2X technologies in the connected environment. Scholars have conducted research on the effectiveness of HUD warnings in pedestrian crossing events outside the zebra crossing (Table 1). The effectiveness of HUD is mainly quantitatively evaluated from visual performance, behavioral performance, and subjective performance. Behavioral performance is considered as direct and beneficial evidence for demonstrating the effectiveness of

Table 1. Summary of relevant research on the application of HUD warning in avoiding pedestrian collisions.

Authors	Objective	Safety-critical events	Methods	Observed variables	Findings in pedestrian events
Liu and Wen, (2004)	Compare the difference of driving performance between HUD and HDD	Vehicles interact with pedestrians on the road outside the zebra crossing	Driving simulation	Brake reaction time	Decreased brake reaction time
Naujoks and Neukum (2014)	Investigate the influence of the specificity and timing of warnings on drivers' driving performance	Vehicles interact with pedestrians on the road outside the zebra crossing ; Vehicles interact with left-turning vehicles at the intersection; Vehicles interact with cyclists at the intersection	Driving simulation	Maximum brake pedal pressure; Minimum Time-to-arrival	Decreased maximum brake pedal pressure; Increased Minimum Time-to-arrival
Winkler et al., (2015)	Investigate the influence of HUD warning on drivers' gaze and driving Behavior	Vehicles interact with pedestrians on the road outside the zebra crossing	Driving simulation	Brake reaction time; Maximum braking value; Number of collisions	Decreased brake reaction time; Increases maximum braking value
Hajiseyedjavadi et al. (2018)	Investigate the effectiveness of visual warnings on young drivers' hazard anticipation and hazard mitigation abilities	Vehicles interact with pedestrians on on roads with and outside zebra crossings; Vehicles interact with vehicles on the road	Driving simulation	Proportion of latent hazards anticipated; Speed	Increased the possibility of glancing at the potential pedestrian; Reduced speed
Winkler et al. (2018a)	Investigate the effectiveness of HUD warnings under different safe-critical events	Vehicles interact with pedestrians on the road outside the zebra crossing and at the intersection; Vehicles interact with bicyclists on the road; Vehicles interact with the lead vehicle on the road ; Vehicles interact with obstacles on the road;	Driving simulation	Brake reaction time; Brake maximum; Collision frequency	Decreased brake reaction time; Increases maximum braking value
Winkler et al. (2018b)	Investigate the influence of multi-stage collision warning on drivers' driving performance	Vehicles interact with pedestrians on the road outside the zebra crossing and at the intersection; Vehicles interact with bicyclists on the road; Vehicles interact with the lead vehicle on the road; Vehicles interact with obstacles on the road;	Driving simulation	Brake reaction time; Brake maximum; Collision frequency	Decreased brake reaction time
Large et al. (2019)	Investigate the effect of urgency and modality of pedestrian alert warnings on driver acceptance and performance	Vehicles interact with pedestrians on the road outside the zebra crossing	Driving simulation	TTC (Gaze on Pedestrian); TTC (Foot off Accelerator); TTC (Foot on Brake); Headway (Car Stopped)	Increased TTC (Foot off Accelerator); Increased TTC (Foot on Brake); Increased Headway (Car Stopped)
Ali, Sharma, et al. (2020)	Investigate the impact of the connected environment on driving behavior and safety	Vehicles interact with pedestrians on the road with zebra crossings	Driving simulation	TTC	Increased TTC
Haque et al. (2021)	Investigate the impact of HUD warnings on latent and observable response time in the connected environment	Vehicles interact with pedestrians on the road with zebra crossings	Driving simulation	Latent response time ; Observable response time	Increased latent response time; Decreased observable response time

TTC is the abbreviation of time-to-collision.

HUD and is currently the focus of scholars' attention. According to the driver's cognitive avoidance process, the perception and decision, manipulation, and risk-avoidance result are the three main aspects of behavioral performance (Dozza et al., 2020; Kudarauskas, 2007; W. Yang et al., 2019).

Perception and decision are the processes in which the driver recognizes the danger and makes a response. It is usually measured accelerator pedal release time and brake reaction time. Liu and Wen (2004) used driving simulation to explore the driving performance of HUD and HDD when completing different tasks. The results of ANOVA indicate that compared to HDD, HUD improves the driver's response speed to pedestrians. Haque et al. (2021) divided the driver response stage into the latent response phase and the observable response phase and used driving simulation to explore the impact of the HUD warning accompanied by beeps on the two response stages in a connected environment under the pedestrian crossing event at zebra crossing. Through T-test, it was found that when providing HUD warnings to drivers, they had a longer latent response time and a shorter observable response time. Large et al. (2019) designed a driving simulation experiment and used the ANOVA method to study the impact of the urgency and modality of pedestrian warning on driving performance. The study found that providing earlier HUD warnings with visual and/or auditory icon helps drivers locate potential threats earlier and encourages them to start deceleration operations earlier.

Manipulation reflects the braking deceleration behavior of the driver to avoid collisions, commonly characterized by maximum deceleration, maximum brake pedal force, and average speed. Hajiseyedjavadi et al. (2018) studied the effectiveness of different warning timing on young drivers hazard anticipation and hazard mitigation abilities through driving simulation. Speed curve analysis revealed that the HUD visual warning group, which received warnings 3 or 4 seconds in advance, maintained significantly lower driving speeds compared to the control group or the 2-second warning group. Naujoks and Neukum (2014) studied the effects of HUD audio-visual warning time and warning specificity (direction specificity and conflict specificity) on driver driving performance through a driving simulator. The results of repeated measurement analysis of variance suggest that providing earlier warning (3.87s) in pedestrian crossing events outside the zebra crossing can reduce the driver's braking intensity.

The risk avoidance result is the ultimate manifestation of the effectiveness of HUD warning on pedestrian collision avoidance, which is often quantified by collision frequency, minimum collision distance, time-to-collision (TTC)/time-to-arrival (TTA) and other indicators. Taking pedestrian crossing events outside the zebra crossing as an example, Winkler et al. (2015) explored the impact of different HUD visual warning contents (warning strategies and specificity) on driver driving performance using a driving simulator. It was found that all driver warnings affected the drivers' performance positively. Although the number of collisions did not decrease, the HUD warning significantly reduced the severity of the collision. Ali et al. (2020) tested the impact of the connected environment on driving behavior and safety in various safety-critical scenarios through driving simulation experiments and conducted data analysis using a linear mixed model (LMM). The results indicate that in a connected environment, HUD pedestrian warnings accompanied by beeps help drivers maintain a longer time-to-collision to the pedestrian.

In fact, the perception and decision, manipulation, and risk avoidance result are not mutually independent. The driver's manipulation behavior will be influenced by the perception and decision, and the final risk avoidance result will be jointly influenced by the perception and decision and manipulation. They collectively reflect the driver's braking avoidance strategy. Previous studies have primarily explored the impact of the HUD warning on driver perception and decision, manipulation, and risk avoidance results from the perspectives of warning modality, timing, and content. However, it is currently unclear how different warning types (HDD and HUD) affect driver braking risk avoidance strategies. The purpose of this paper is to explore how different warning types (HDD and HUD) affect driver braking risk avoidance strategies and whether the HUD warning can help drivers choose the optimal braking risk avoidance strategy.

2.3. Test tool for HUD warning utility

From Chapter 2.2, it can be seen that driving simulation has become an important tool for testing the effectiveness of HUD warnings. It allows to construct a connected environment to realize effective interaction and real-time warning among vehicle-to-vehicle, vehicle-to-cyclist and vehicle-to-pedestrian (Ali et al., 2020). At the same time, it can create safety-critical scenarios and various weather conditions, allowing experimenters to observe user behavior in safe environments without worrying about any accident costs (Calvi et al., 2020). In addition, driving simulators can provide a high degree of experimental control and fine-grained driving behavior data, which has been widely used in micro-driving behavior analysis (Raddaoui et al., 2020; Zhao et al., 2021). It is undeniable that driving simulations also have certain limitations. Compared to real driving, the driver's psychological feelings and perception of speed and distance information in driving simulation may be different (Ross et al., 2021). It is worth noting that although driving simulations have inherent limitations, relative validity is still feasible with a driving simulator (Fisher et al., 2011; Galante et al., 2010), which was confirmed by recent work by Hussain et al. (2019) with respect to speed perception and actual speed. The braking behavior of drivers in driving simulators has also been proven to be comparable to real driving (Boda et al., 2018). Previous studies on the effectiveness testing of warning systems have also cited relevant literature to theoretically demonstrate that the limitations of driving simulations do not affect the effectiveness of research results (Brijs et al., 2022; Winkler et al., 2018a). Based on this, driving simulation is used as the testing tool for this study. By using a highly simulated driving simulator and constructing a more realistic road environment, this paper provides drivers with a more realistic driving experience, further narrowing the difference between driving simulation and the real environment.

2.4 A method of exploring multiple influencing factors

Similar to general linear models and generalized linear models, the generalized linear mixed model (GLMM) is also a commonly used method for exploring influencing factors and quantifying the degree of impact. The generalized linear mixed model (GLMM) combines the characteristics of two statistical frameworks: the linear mixed model (random effects) and the generalized linear model (non-normal data), which can deal with a variety of research designs and data types, and fit data with non-normal distributions and complex correlation structures (Fei et al., 2013; Nelder & Wedderburn, 1972). GLMM has been widely applied in driving behavior research. Zhao et al. (2022) used GLMM to explore the influence of driver attribute factors and takeover situation factors and their interaction on takeover behavior. Adavikottu and Velaga (2023) quantitatively analyzed the impact of factors such as driver characteristics and different levels of aggressive driver behavior on the collision risk at unsignalized intersections using GLMM. Based on this, considering the data structure and distribution, this paper selects GLMM to quantitatively analyze the factors that affect the driver's braking avoidance strategy.

3. Methods

Based on a driving simulator, this paper has completed the construction of the connected experimental scenario and the development of HUD and HDD interfaces.

All subjects must drive in six urban road scenarios: three warning display types (Baseline/HDD/HUD) combined with two weather conditions (clear weather/foggy weather). Then, based on fine-grained driving behavior data, the risk avoidance process was characterized and indicators were extracted. Finally, considering the data structure and distribution of this paper, the Generalized Linear Mixed Model (GLMM) was used to explore the factors affecting brake risk-avoidance strategies.

3.1. Connected driving simulation experiment platform

This paper builds a driving simulation experiment platform under the connected environment, which is composed of the driving simulation system, the human factor data-acquisition system, the data collaborative processing center and the vehicle terminal (human machine interface, HMI). Each component is interconnected using wireless communication technology. The architecture of the experimental platform is shown in Figure 1.

3.1.1. Driving simulation system

The driving simulation system is composed of a fixed-base driving simulator and a computer located in Beijing University of Technology, as shown in Figure 2. The driving simulation system has the functions of virtual scenario design and development, on-board data acquisition, data transmission, etc. The design and development of virtual scenarios include the construction of roads and the setting of built-in parameters. After completion, the scenario is imported into the driving simulation system, and the weather, traffic flow and other scenario elements are added through the application program interface (API). The on-board data acquisition includes the operating parameters of the vehicle and adjacent related vehicles (such as speed, acceleration, brake-pedal depth, accelerator-pedal efficacy,

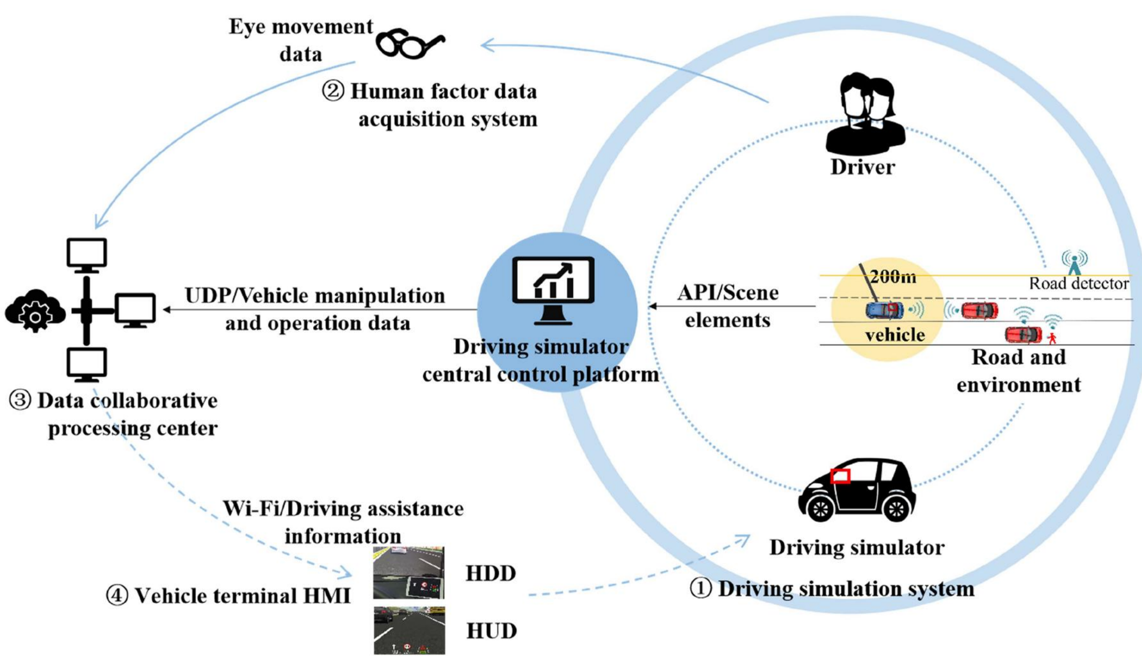


Figure 1. Connected driving simulation experiment platform.

lateral position, etc.). The acquisition frequency is set to 30 Hz and transmitted to the collaborative data management center through the user data protocol (UDP).

3.1.2. Human factor data-acquisition system

The human factor data-acquisition system obtains the drivers' eye-movement data through the eye-movement instrument. This paper only analyzes driving behavior, and the eye-movement data obtained will be studied in other articles.

3.1.3. Data collaborative processing center

The data collaborative processing center receives the vehicle, road and environment information from the driving simulation system in real time, and the transmission mode can be regarded as the combination of V2V and V2I communication. This information further filters and fuses the data; and then interconnects with the vehicle terminal HMI over WiFi.

3.1.4. Vehicle terminal HMI

The vehicle terminal HMI involved in this paper was one of either HDD or HUD. The HDD or HUD receives the filtered and fused data from the data collaborative processing center and completes the real-time dynamic release of driving assistance information.

3.2. Experimental design

3.2.1 Information interaction design

The experimental design included three warning display types, which were: Baseline, HDD and HUD. Only voice navigation information was provided in the Baseline condition, while HUD and HDD provide safety warning information in the connected environment.

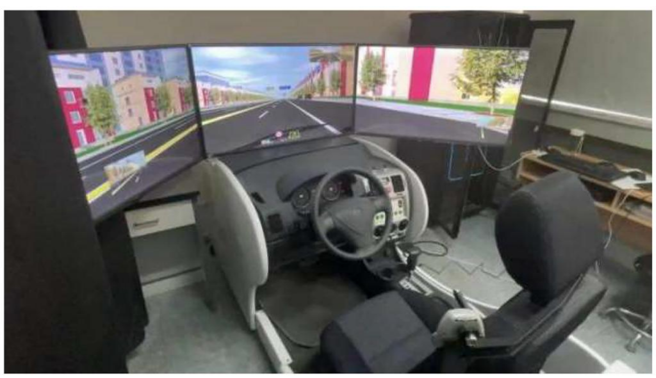


Figure 2. Fixed-base driving simulator.

In the interface design of HUD, this paper comprehensively considered the cognitive ability of the driver, the size limitation of the HUD (Strayer et al., 2019) and the current situation of in-vehicle driving assistance system products (Audi A7). At the same time, the interface design principles of importance, use frequency, functional grouping and use sequence were followed (Abdel-Aty et al., 2007). The HUD interface in the connected environment was finally designed and developed as shown in Figure 3. According to the contents of the information to be displayed, the HUD was divided into four sections: vehicle state information, navigation information, safety warning information and entertainment and communication information. The safety warning information section can present continuous and on-time event-triggered connected information, such as continuous information about the spacing to the leader and event-triggered information about pedestrian crossing. The displayed contents of each section are shown in Figure 3.

The size of the HUD is greatly affected by the type of vehicle and interface design, and there is no unified standard. Taking into consideration the type of driving simulator and the consistency with HDD size, we set the size of the HUD interface at 215.5 mm × 124.2 mm. The HUD was developed using the SCANeR software, which was integrated into the driving simulation, used WiFi real-time communication, and was finally displayed on the road above the steering wheel. In order to ensure the uniqueness of variables, the interface design and size of the HDD were completely consistent with those of the HUD. The only difference was the position of the two displays. HDD was displayed by HUAWEI pad M3 and placed in the right front of the vehicle steering wheel, which was similar to the position of the central control panel in the vehicle. See Figure 4.

3.2.2 Scenario design

Six scenarios were designed for the experiment, covering three warning display types (Baseline/HDD/HUD) combined with two weather conditions (clear weather/foggy weather). When using a warning display, the driver's experiences from start to finish are shown in Figure 5(a). The three warning display types were connected into a loop, and the driver drove a circle in each of the two weather conditions (clear

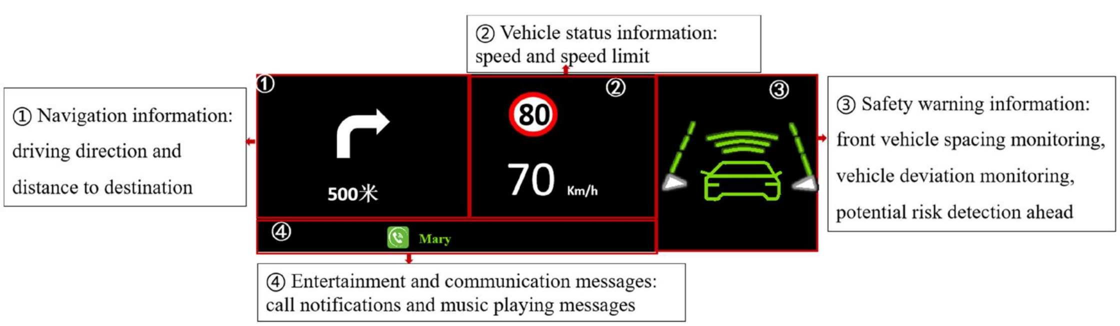


Figure 3. HDD/HUD interface design.

(a) HDD position (b) HUD position

Figure 4. HDD/HUD setting position.

weather/foggy weather). Except for the warning display type and weather, the event design in all six scenarios is the same. To avoid driving fatigue and learning effects, multiple bus stops and roadside parked buses will be set up on the road sections outside the scope of the event's impact in each scenario. Pedestrians or animals may appear behind some parked buses. And we will change the color, number and type of parked vehicles on the roadside. In addition, the surrounding environment (buildings, trees) also changes in each scenario.

The experimental scenario was a 4 km long urban road, and five typical, hazardous events are designed, including left-side vehicle merging, conflict with traffic in the opposite direction, hidden pedestrian crossing outside the zebra crossing, and conflict with turning left and emergency stop of vehicle ahead, as shown in Figure 5(a). This paper only considers the hidden pedestrian crossing event outside the zebra crossing; the rest will be studied in other articles.

The scenario design of the hidden pedestrian crossing event outside the zebra crossing is shown in Figure 5(b). The road where the event occurs is a two-way, four lane road with a lane width of 3.5 m, a sidewalk width of 5 m and a speed limit of 60 km/h. The experiment set up two weather conditions: clear weather, and foggy weather with visibility of 240 m. According to the "Grade of fog forecast" (GB/T 27964 – 2011, China), the visibility of dense fog condition is less than 500 m and more than 200 m. This visibility fog reduces the driver's perception of the surrounding environment and speed, which can affect their decision and behavior (Abdel-Aty et al., 2011; Yan et al., 2014). In the free-flow state, there was a stable traffic flow in the left lane of this road section, forcing the main vehicle to drive only in the right lane, and there were no other vehicles ahead in the main vehicle's lane, so the longitudinal behavior of the main vehicle was not constrained. The pedestrian was 2.5 m away from the right edge line of the lane and was blocked by the bus. In order to avoid learning effects as much as possible, the type and color of buses will be changed during scenario design, and some parked buses will be added on the roadside, but no pedestrians will appear. The roadside equipment/infrastructure detected the pedestrian (a typical V2I case), and both the HUD and HDD in the connected environment started issuing warning messages when the main vehicle reached the point 100 m from the pedestrian. When the main vehicle reached the point 60 m from the pedestrian, the pedestrian was activated and began to cross the road at a speed of 1 m/s. The main vehicle could only slow down to avoid pedestrians, and the event ended after it passed the pedestrian position.

The HDD and HUD warnings designed in this paper are accompanied by voice prompts. The voice prompt "pay attention to pedestrians in front" started at 100 m from the pedestrians, and the prompt was given only once. At the same time, the safety warning section in the HUD and HDD interface indicated the graphic symbol for pedestrians crossing the road in front, and the warning graphic symbol did not disappear until the vehicle passed the pedestrian position, as shown in Figure 5(c).

3.3. Participants

In this experiment, subjects from universities and social groups who met the requirements were recruited. The recruited subjects needed to meet three conditions: (1) have or obtain a motor vehicle driver's license; (2) normal hearing and vision; and (3) no nausea, dizziness or other problems. After screening, forty eligible subjects were recruited, including five pre-experimenters and thirty-five formal experimenters. The main purpose of the pre-experiment was to test the rationality of the experimental design and arrangement, so statistical analyses were not done. One of the formal experimenters withdrew from the experiment due to scheduling conflicts, so the final sample size was thirty-four, of which twenty-five (73.53%) were male and nine (26.47%) were female. The sex ratio was roughly in line with current statistical characteristics of Chinese drivers (Ministry of Public Security of the People's Republic of China, 2020). The subjects' age is between 21 and 53 years (Mean = 29.21, SD = 9.46). Their driving experience ranges between 1 and 22 years (Mean = 6, SD = 5.16). The demographic data for the thirty-four subjects in the formal experiment are shown in Table 2.

In order to ensure the effectiveness of the results, it is necessary to calculate the sample size required for the experiment. The required sample size was calculated according to the expected variance, target confidence level and error amplitude (Chen et al., 2024; Zhao et al., 2021):

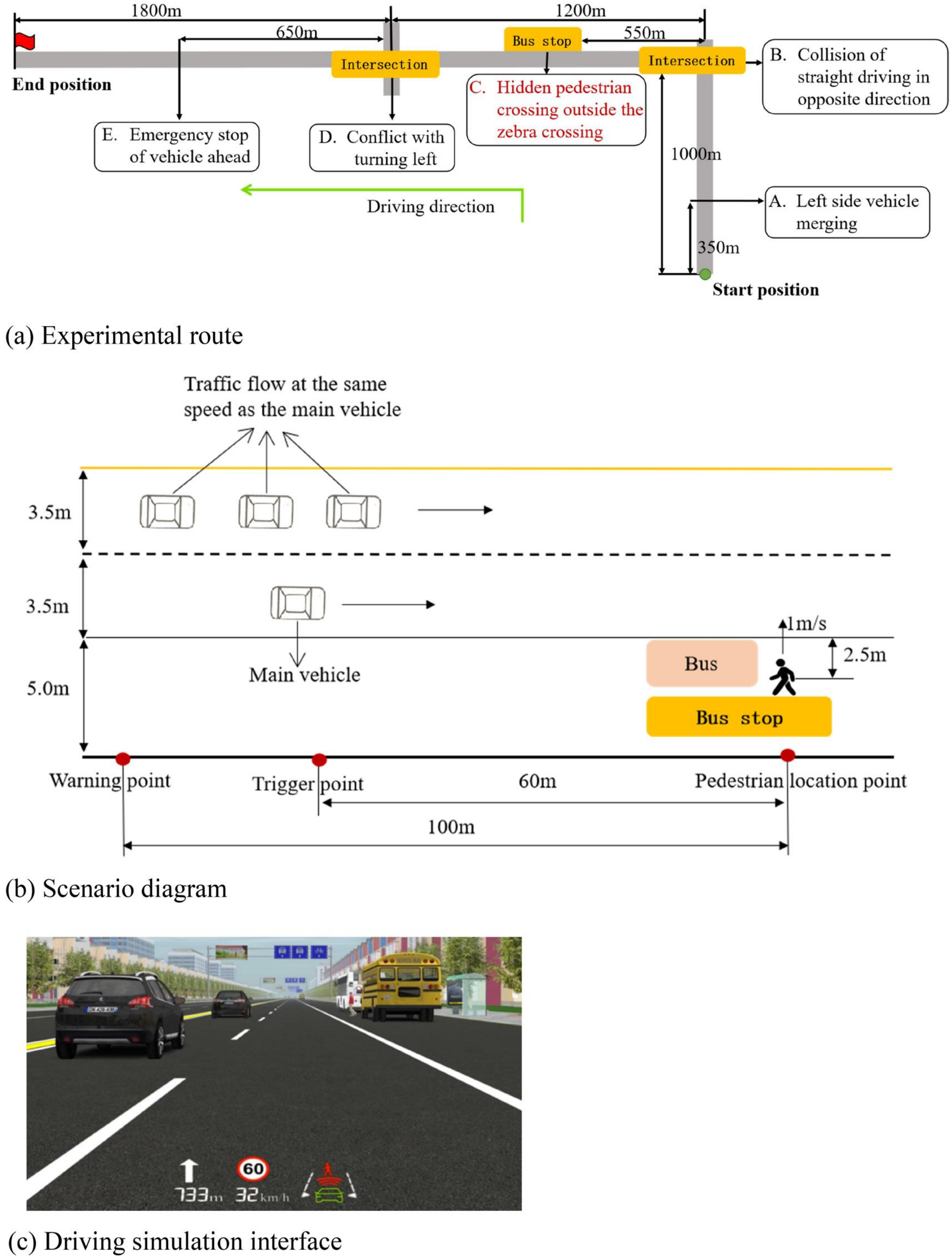


Figure 5. Scenario design of the hidden pedestrian crossing event outside the zebra crossing.

$$N = \frac{(z_{\alpha/2} + z_\beta)^2\sigma^2}{\varepsilon^2}$$

where N is the required sample size, $z_{\alpha/2}$ is the upper $(\alpha/2)$th quantile of the standard normal distribution; z_β is the upper (β)th quantile of the standard normal distribution; σ is the standard deviation of the normal distribution population; ε is the difference between the true mean response of a test factor and a reference value, which can be given by $\varepsilon = \pm\delta\sigma$ (Lerman, 1996), δ is the meaningful difference, in practice, a value between 0.25 and 0.5 is usually chosen as δ if no prior knowledge (Chow et al., 2017). Typically, a 5% level of significance is chosen to reflect a 95% confidence regarding the unknown parameter, $z_{\alpha/2}$=1.96. To balance the power and cost-effectiveness, a power of 80% and a meaningful difference of 0.5 were used, z_β=0.84, δ=0.5. The results showed that the required sample size in this research was 32, and the number of participants in this study was 34. So the sample size selected in this experiment met the requirements, and could provide reliable answers to the research questions.

Table 2. Statistics of driver's basic information.

Attribute	Level	Number	Ratio (%)
Gender	Male	25	73.53
	Female	9	26.47
Age	< 26	16	47.06
	26-–57	18	52.94
Driving experience	Beginner (< 3 years)	16	47.06
	Middle (4–9 years)	11	32.35
	Senior (> 10 years)	7	20.59
Driving frequency	Rarely driving (Driving less than four times a month)	20	58.82
	Driving every week	14	41.18
Prior experience with HUD	Not at all	15	44.12
	Heard it before, but wasn't sure what it was.	10	29.41
	Know but never used	5	14.71
	Used	4	11.76

3.4. Experimental procedure

There were six formal scenarios in the experiment (see Section 3.2). Each subject was asked to drive in all six scenarios to ensure a similar distribution of subject characteristics in the experiment. In order to eliminate the memory effect as much as possible, the scenario orders for each participant were determined using a Latin square. In addition, the six scenarios were arranged to be completed over two working days, with an interval of three days in between. For scenarios completed on the same day, subjects were required to rest for five minutes after each scenario was completed, and will be arranged to complete another city road scenario that is not used for analysis. There are a total of six such informal scenarios, with three assigned per workday. The specific experimental steps were as follows:

Before the test: subjects filled in a demographic questionnaire, recording gender, age, driving experience and other information; the experimenter briefly introduced the experimental process and the on-board information interface to the subjects; the subjects were asked to complete a pre-test in an informal test scenario to test whether the subjects adapted to the driving simulator environment; the experimenters read out the experimental instructions to the subjects, and told them to follow their regular driving habits, abide by the traffic rules and information guidance, and complete the driving task safely.

In the test: subjects adjusted their seats to sit in a proper position; subjects conducted the driving simulation experiment in Latin square order, with an individual experimental scenario lasting about five minutes; after a single scenario, subjects rested for approximate five minutes; on the first working day, subjects repeat steps – to complete the set of three formal experimental scenarios and three informal experimental scenarios not used for analysis; after three days, the remaining scenarios were completed in the second working day, and experimental steps – repeated.

After the test: the experimenters checked the behavioral data records of the subjects. After verifying that there were no missing data, each subject received a reward of 160 yuan.

3.5. Data extraction

For the hidden pedestrian crossing event outside the zebra crossing, the warning point and pedestrian position are the start and end points of the impact range for the event, respectively; this was a length of 100 m, as shown in Figure 5(b). Basic driving behavior data were collected within these 100 m, including speed, acceleration, brake-pedal depth and accelerator-pedal efficiency, etc. The required indicators can be further calculated from basic driving behavior data.

3.6. Process analysis and index extraction

3.6.1. Description of braking risk-avoidance process

In the experimental design of this paper, drivers can only avoid danger by braking. Thus, this paper provides a detailed description of the braking risk-avoidance process for drivers under the hidden pedestrian crossing event outside the zebra crossing. In the scenario of using an HUD in clear weather, we selected one driver from 34 drivers and drew a spatial variation diagram of his driving behavior within 100 m from pedestrians, as shown in Figure 6. In this figure, the ordinates are speed, acceleration, brake-pedal efficiency and accelerator-pedal efficiency, respectively; the abscissa is the distance, where 0 m is the warning point of the HUD and HDD, 40 m is the pedestrian trigger point, and 100 m is the pedestrian point. Figure 6 contains two dimensions: vertical and horizontal.

3.6.1.1. Vertical dimension. According to the running characteristics of the vehicle, the vertical dimension is divided into three levels: (i) manipulation, (ii) operation and (iii) risk avoidance, and these three levels are progressive. (i) The manipulation level reflects the manipulation and control behavior of the vehicle, including the spatial-change curve of brake-pedal efficiency and accelerator-pedal efficiency; (ii) the operation level reflects the operation status of the vehicle, including the spatial-change curve of speed and acceleration; (iii) the risk-avoidance level reflects the driver's cognitive risk-avoidance process, including the start point of risk avoidance (warning point) and the end point of risk avoidance (minimum collision distance point).

3.6.1.2. Horizontal dimension. According to the cognitive processes of drivers, the horizontal dimension is divided into three stages: (i) perception and decision, (ii) risk-avoidance manipulation and (iii) risk-avoidance result. (i) The perception and decision stage reflects the driver's perception-judgment-decision process, starting from the warning

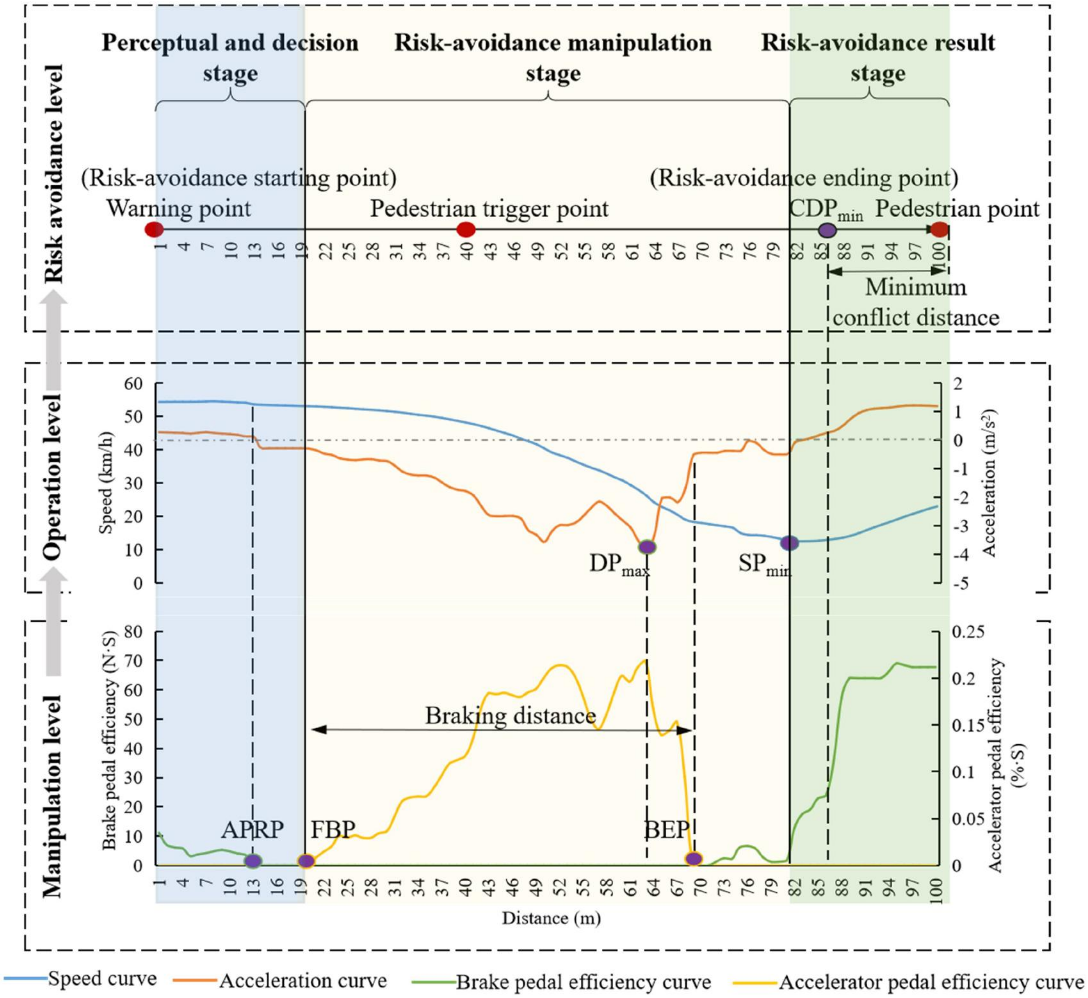


Figure 6. Spatial variation of braking risk-avoidance processes in HUD group under clear weather conditions. (full name of abbreviations in the figure: accelerator-pedal release point (APRP), first braking point (FBP), maximum deceleration point (DPmax), brake end point (BEP), minimum speed point (SPmin), minimum collision distance point (CDPmin).

point (the risk-avoidance starting point) and ending at the point of the first braking. The key points include the accelerator-pedal release point and the first braking point. (ii) The risk-avoidance manipulation stage reflects the process in which the driver effects the deceleration through the manipulation and adjustment of the brake pedal, starting at the first braking point and ending at the minimum speed point. The key points include the maximum deceleration point and the braking end point. (iii) The risk-avoidance result stage reflects the recovery process after the deceleration is completed, starting at the minimum speed point and ending at the pedestrian point. The key points include the minimum speed point and the minimum conflict distance point (the risk-avoidance ending point). Based on the detailed description of the whole braking risk-avoidance process and the marking of key points in space, the braking risk-avoidance strategy can be obtained.

In order to verify that the key points we selected can reflect different braking risk-avoidance processes and strategies, we further characterized the same driver's spatial variation of braking risk-avoidance process in the Baseline, HDD and HUD groups under clear weather conditions, as shown in Figure 7. Figure 7 marked the key points selected in Figure 6, including the accelerator-pedal release point, the first braking point, the maximum deceleration point, the braking end point, the minimum speed point, and the minimum conflict distance point. It can be seen that there are differences in the key points among the Baseline, HDD and HUD groups.

3.6.2. Index extraction

Based on the description of the process in the previous section, we selected the corresponding indicators from the three stages of perception and decision; risk-avoidance manipulation; and risk-avoidance results to reflect the driver's braking risk-avoidance strategy (see Table 3).

3.7. Model selection and construction

3.7.1. Model selection

This paper aims to explore the effects of three warning display types (Baseline/HDD/HUD) combined with two weather conditions (clear weather/foggy weather) on drivers' braking risk-avoidance strategies; consider differences between drivers' attributes (gender, driving experience,

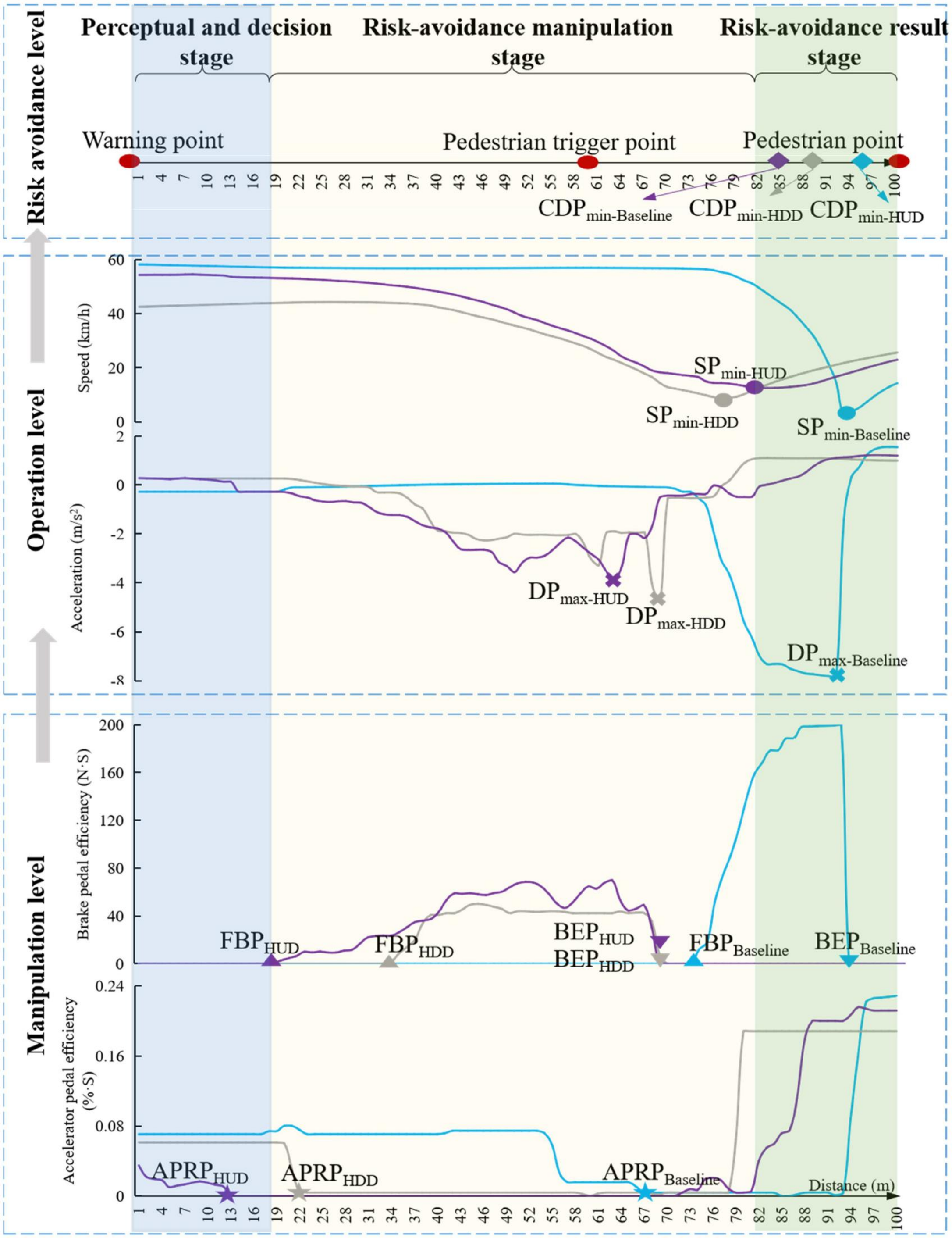


Figure 7. Spatial variation of braking risk-avoidance processes in baseline, HDD and HUD groups under clear weather conditions. (note that the blue lines and symbols represent the baseline group; grey lines and symbols represent the HDD group; purple lines and symbols represent the HUD group.) (the full name of the abbreviations are consistent with those in Figure 6, and baseline, HDD and HUD of subscript forms combined with abbreviations respectively.).

driving frequency); and the interaction among these different factors. In order to explore the factors affecting the indicators of the braking risk-avoidance strategy and quantify the differences between levels of the same factor, taking into account the distribution of the data (the Shapiro-Wilk test showed that some of the experimental data do not follow the normal distribution) and the random effects caused by repeated experiments and differences between individuals, a Generalized Linear Mixed Model (GLMM) was selected.

3.7.2. GLMM construction

Taking the six indicators extracted in Section 3.6 as dependent variables, six GLMMs were established. The independent variables included fixed effects and random effects. The fixed effects included warning display types, weather conditions, and driver attributes (gender, driving experience, driving frequency). The random effects included: (i) correlation of repeated measurement data (warning display types, weather conditions); and (ii) individual heterogeneity (gender,

Table 3. Index system.

Braking risk-avoidance strategy	Indexes	Description	Notes
Perception and decision stage	Position of accelerator-pedal release (m)	Position of accelerator pedal release is the distance between the driver's first accelerator pedal release point and the warning point	The driver senses the pedestrians in front through warnings or observation. In order to avoid a collision, the driver releases the accelerator pedal, which represents the driver's perception stage.
	Position of first braking (m)	Position of first brake is the distance between the driver's first braking point and the warning point.	After perception takes place, the driver makes a preliminary judgment according to the current environment and the operational status of the vehicle. Stepping on the brake is the key point between judgment and action.
Risk-avoidance manipulation stage	Maximum deceleration (m/s^2)	Maximum deceleration is the maximum deceleration taken by the driver within the 100 m between the warning point and the pedestrian position.	The maximum deceleration is used to reflect the urgency of the driver in the face of hazard and the efforts that the driver needs to make to avoid a collision.
	Braking distance (m)	The braking distance is the distance that the driver passed after stepping on the brake pedal within the 100 m between the warning point and the pedestrian position.	The braking distance combined with the maximum deceleration is used to reflect the urgency of the driver in the face of hazard and the stability of the deceleration control process.
Risk-avoidance result stage	Minimum collision distance (m)	Minimum collision distance is the minimum collision distance between the main vehicle and the pedestrian once the pedestrian starts moving, until they leave the lane where the main vehicle is located.	The minimum collision distance reflects the final risk-avoidance result.
	Position of minimum speed (m)	Minimum speed position is the distance between the position where the driver reaches the minimum speed and the warning point.	The driver slows down, reduces the speed to the lowest point, and then accelerates again. The position of minimum speed reflects the end point of risk-avoidance in the driver's mind.

Table 4. Model construction.

Effect of type	Factor		Level
Fixed/Random effects	Warning display types		Baseline/HDD/HUD
	Weather conditions		clear weather/foggy weather
	Driver attributes	Gender	Male/Female
		Driving experience	Beginner (<3 years) Middle (4–9 years) Senior (>10 years)
		Driving frequency	Driving every week/ Rarely driving
Dependent variables	Position of accelerator pedal release、Position of first brake、Maximum deceleration、Braking distance、Minimum collision distance、Position of minimum speed		

driving experience, driving frequency). The specific model construction is included in Table 4.

4. Results and discussion

In this chapter, descriptive statistics were completed for six indicators under different warning display types and weather conditions. And a detailed analysis and discussion were conducted on the obtained GLMM results. Finally, the limitations and future research of this paper were explained.

In order to ensure the effectiveness of the results, the impact of the learning effect was separately tested. According to the experimental order, all data were divided into 6 groups. Considering that the data does not follow a normal distribution (Shapiro-Wilk test) and the correlation between samples, the Friedman test in the non parametric test was selected, and the test results are shown in Table 5. The results showed that there was no significant difference in each indicator among different experimental order groups ($p > 0.05$), indicating that the learning effect did not reach a significant impact.

4.1. Descriptive statistics

Descriptive statistics were calculated for the six indicators under different warning display type and weather conditions, and the results were expressed in the form of mean value (standard deviation, SD), as shown in Table 6. Scatter plots of the six indicators for all thirty-four drivers using the different warning display types are also shown. It can be seen from the plots that there may be differences between indicators under different warning display types and weather conditions.

4.2. Model result analysis and discussion

First, the data was pre-processed to delete abnormal or wrong data according to Pauta standard, also referred to as 3σ criterion (Du et al., 2021). Based on the processed data, the GLMM was established, and the model was fit with the help of SPSS 25. Then, the covariance structure with the minimum Akaike information criterion (AIC) and Bayesian information criterion (BIC) was selected. The fixed effects and fixed-effect solutions for the model are shown in Tables 7 and 8.

Table 5. Results of learning effect test.

Braking risk-avoidance strategy	Indexes	P
Perception and decision stage	Position of accelerator-pedal release (m)	0.556
	Position of first braking (m)	0.595
Risk-avoidance manipulation stage	Maximum deceleration (m/s2)	0.581
	Braking distance (m)	0.696
Risk-avoidance result stage	Minimum collision distance (m)	0.331
	Position of minimum speed (m)	0.054

Table 6. Descriptive statistics.

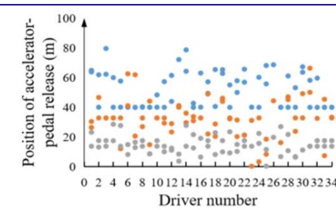
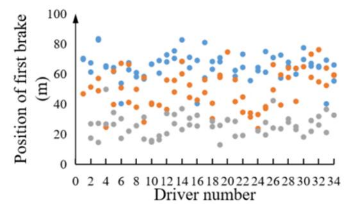
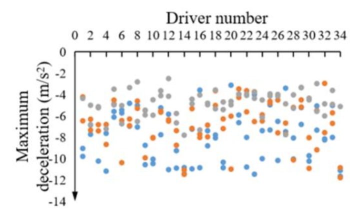
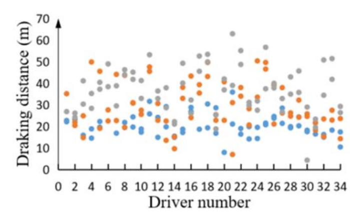
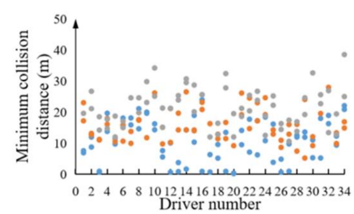
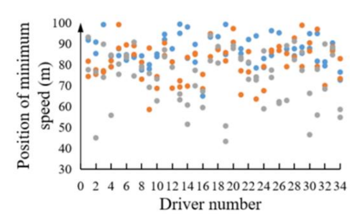
		Mean (SD)		Scatter diagram legend: Baseline group, HDD group, HUD group
Indexes	Warning display types	Clear weather	Foggy weather	
Position of accelerator-pedal release (m)	Baseline	49.99 (11.53)	55.02 (15.46)	
	HDD	32.02 (9.87)	32.50 (15.33)	
	HUD	16.10 (6.81)	13.34 (4.29)	
Position of first brake (m)	Baseline	64.50 (10.22)	68.22 (10.07)	
	HDD	53.16 (12.26)	47.18 (12.73)	
	HUD	27.33 (7.98)	23.96 (5.72)	
Maximum deceleration (m/s^2)	Baseline	−8.30 (1.94)	−7.77 (2.66)	
	HDD	−6.94 (2.34)	−6.59 (1.88)	
	HUD	−5.03 (1.23)	−4.61 (1.12)	
Braking distance (m)	Baseline	21.24 (5.12)	19.55 (5.09)	
	HDD	27.00 (10.60)	31.60 (10.61)	
	HUD	36.83 (8.08)	36.73 (12.96)	
Minimum collision distance (m)	Baseline	10.75 (6.97)	10.63 (6.42)	
	HDD	15.02 (5.25)	16.36 (5.38)	
	HUD	20.96 (6.58)	21.65 (5.91)	
Position of minimum speed (m)	Baseline	86.14 (7.85)	86.94 (5.86)	
	HDD	81.24 (9.26)	81.42 (8.41)	
	HUD	73.75 (12.64)	74.52 (13.20)	

Table 7 only shows the variables that had significant effects on the six indicators of a driver's braking risk-avoidance strategy. First, in the two indicators for the perception and decision stage, only the warning display type ($p < 0.001$) had a statistical difference on the position of accelerator-pedal release; warning display type ($p < 0.001$), driving frequency ($p = 0.019$), interaction between gender and driving experience ($p = 0.002$), and interaction between driving experience and driving frequency ($p = 0.009$) had significant effects on the position of first braking. Second, in the two indicators for the risk-avoidance manipulation stage, there were statistical differences in the effects of warning display type ($p < 0.001$), weather conditions ($p = 0.016$), driving experience ($p = 0.009$), interaction between weather conditions and gender ($p = 0.026$), and interaction between driving experience and driving frequency ($p = 0.008$) on maximum deceleration; the effects of warning display type ($p < 0.001$) and the interaction between warning display type and gender ($p = 0.021$) on braking distance were statistically different. Finally, in the two indicators for the risk-avoidance result stage, the impacts of warning display type ($p < 0.001$), driving experience ($p = 0.017$), driving frequency ($p = 0.020$), and the interaction between gender and driving frequency ($p = 0.037$) on the minimum conflict distance were statistically different; only warning display type ($p < 0.001$) had a significant effect on the position of minimum speed.

Table 8 only shows the variables that had significant effects on the six indicators and gives the coefficients of their fixed effect solutions. The difference between different levels of the same independent variable is then obtained. The analysis of Table 8 is divided into analysis of indicator characteristics and analysis of influencing factors.

4.2.1. Analysis of indicator characteristics

Analysis of indicator characteristics refers to analyzing the influencing factors of six indicators and the differences between different levels of the same factor from three stages in Table 8.

4.2.1.1. Perception and decision stage

For the position of accelerator-pedal release and the position of first braking, smaller values indicated closer proximity to the starting position, which meant that the driver completed the perception and decision stage earlier (Calvi et al., 2020).

4.2.1.1.1. Position of accelerator-pedal release. Taking the Baseline group as the reference, the position of accelerator-pedal release for the HUD and HDD groups were significantly closer to the starting position (HUD: $p < 0.001$; HDD: $p < 0.001$); of these, the HUD group was closer. Compared to drivers with beginner driving experience (< 3 years), drivers with middle driving experience (4–9 years) released the accelerator pedal earlier ($p = 0.030$). Taking the Baseline group as the reference, the interaction between the warning display types and weather conditions showed that the HUD group released the accelerator pedal earlier in foggy weather than in clear weather ($p = 0.032$); the interaction between warning display types and driving experience showed that the position of accelerator-pedal release for drivers with middle driving experience was farther away than that of drivers with beginner driving experience in the HDD and HUD group (HDD: $p = 0.019$; HUD: 0.076). This indicated that HUD and HDD increased the driving confidence of drivers with middle driving experience, and they chose to release the accelerator pedal later. In other words, the improvement of driving confidence also reduced their compliance with HUD and HDD.

4.2.1.1.2. Position of first brake. Taking the Baseline group as the reference, the position of first braking for the HUD and HDD groups was closer to the starting position (HUD: $p < 0.001$; HDD: $p < 0.001$) of these, the HUD group was closer. In terms of driver attributes, drivers with senior driving experience (> 10 years) started braking further away from the starting point, as compared to drivers with beginner driving experience ($p = 0.034$); drivers who drive every week started braking further away than drivers who rarely drive ($p = 0.033$). The reason for this may be that drivers with different driving experiences have different observation strategies (Crundall et al., 1999). After receiving a warning or perceiving the hazard, experienced drivers will spend more time observing the changes in the hazard to determine when to brake (Yeung & Wong, 2015). An interesting finding is that the distance required to complete the perception and decision stages first decreases, and then increases with increasing driving experience. Further exploration is needed to confirm this pattern. Taking the Baseline group as a reference, the interaction between the warning display types and weather conditions showed that the position of first braking of the HUD and HDD groups in foggy weather was earlier than in clear weather (HUD in foggy weather vs. HUD in clear weather, $p = 0.035$; HDD in foggy weather vs. HDD in clear weather, $p = 0.011$). This phenomenon was due to the fact that drivers' ability to perceive the outside environment decreases in foggy weathers, and they rely more on the in-vehicle warning displays, showing higher attention and compliance to the warning information (Yue et al., 2018; Zhao et al., 2021b). The interaction between warning display types and driving experience showed that the position of first braking for drivers with middle driving experience was further away than drivers with beginner driving experience in the HDD group ($p = 0.042$). In the interaction between gender and driving experience, taking male as the reference, the position of first braking of females with senior driving experience was earlier than females with beginner driving experience ($p < 0.001$). In the interaction between driving frequency and driving experience, taking drivers who rarely drive as the reference, drivers with senior driving experience (>10 years) showed an earlier first braking position than those with beginner driving experience (<3 years) among drivers who drive every week (> 10 years) ($p = 0.004$).

4.2.1.2. Risk-avoidance manipulation stage

Smaller absolute values of maximum deceleration and greater braking distances indicated a lower degree of urgency in the manipulation stages (Winkler et al., 2018b).

Table 7. Fixed effect.

Indexes		Perceptual and decision stage		Risk-avoidance manipulation stage		Risk-avoidance result stage	
		Position of accelerator pedal release (m) F(P)	Position of first braking (m) F(P)	Maximum deceleration (m/s^2) F(P)	Braking distance (m) F(P)	Minimum collision distance (m) F(P)	Position of minimum speed (m) F(P)
Warning display types		**164.732** **(<0.001)****	**170.244** **(<0.001)****	**40.513** **(<0.001)****	**30.513** **(<0.001)****	**32.983** **(<0.001)****	**23.157** **(<0.001)****
Weather conditions		0.194 (0.660)	1.285 (0.259)	**5.956** **(0.016)****	0.784 (0.377)	0.022 (0.882)	0.365 (0.547)
Driver attributes	Gender	1.198 (0.275)	**3.496** **(0.063)***	0.406 (0.525)	**2.966** **(0.087)***	0.838 (0.361)	1.059 (0.305)
	Driving experience	0.116 (0.890)	**2.483** **(0.087)***	**4.864** **(0.009)****	0.701 (0.498)	**4.206** **(0.017)****	1.549 (0.216)
	Driving frequency	0.283 (0.595)	**5.584** **(0.019)****	0.345 (0.558)	0.638 (0.426)	**5.475** **(0.020)****	0.016 (0.901)
Interaction effect	Warning display types *Weather conditions	**2.580** **(0.079)***	**3.877** **(0.023)****	0.046 (0.955)	2.188 (0.115)	0.164 (0.849)	0.024 (0.977)
	Warning display types *Gender	1.042 (0.355)	0.665 (0.516)	0.230 (0.795)	**3.968** **(0.021)****	0.569 (0.567)	**2.833** **(0.062)***
	Weather conditions * Gender	0.166 (0.684)	0.148 (0.701)	**5.054** **(0.026)****	0.053 (0.819)	0.077 (0.782)	0.366 (0.546)
	Gender *Driving experience	1.086 (0.340)	**6.417** **(0.002)****	1.723 (0.182)	0.267 (0.766)	**2.607** **(0.077)***	**2.348** **(0.099)***
	Gender *Driving frequency	0.376 (0.540)	**3.878** **(0.051)***	1.204 (0.274)	0.790 (0.376)	**4.442** **(0.037)****	0.437 (0.509)
	Driving frequency *Driving experience	1.468 (0.233)	**4.871** **(0.009)****	**4.905** **(0.008)****	1.776 (0.173)	0.446 (0.641)	0.166 (0.847)
Covariance type		Variance component	First order Autoregression	Variance component	Variance component	Variance component	Variance component
AIC		1364.028	1246.825	753.870	1267.496	1150.048	1310.295
BIC		1391.185	1273.041	783.844	1297.126	1179.679	1340.065

*Significant at 10% (bold values).
**Significant at the 5% (bold values).

Table 8. Model fixed effect solution.

Indexes	Level	Perceptual and decision stage: Position of accelerator pedal release Coefficient (m) (P)	Perceptual and decision stage: Position of first braking Coefficient (m) (P)	Risk-avoidance manipulation stage: Maximum deceleration Coefficient (m/s^2) (P)	Risk-avoidance manipulation stage: Braking distance Coefficient (m) (P)	Risk-avoidance result stage: Minimum collision distance Coefficient (m) (P)	Risk-avoidance result stage: Position of minimum speed Coefficient (m) (P)
Intercept		51.906 (<0.001)**	62.067 (<0.001)**	−7.683 (<0.001)**	21.761 (<0.001)**	9.430 (<0.001)**	86.848 (<0.001)**
Warning display types	HUD (Baseline)	−36.292 (<0.001)**	−39.104 (<0.001)**	3.451 (<0.001)**	19.595 (<0.001)**	9.412 (<0.001)**	−9.194 (0.018)**
	HDD (Baseline)	−23.822 (<0.001)**	−15.242 (<0.001)**	1.832 (0.018)**	8.268 (0.008)**	6.186 (0.007)**	−6.366 (0.042)**
Driving experience	Senior (>10 years) (Beginner (<3 years))	1.782 (0.734)	10.228 (0.034)**	−2.283 (0.012)**	−6.683 (0.030)**	2.446 (0.418)	−0.246 (0.949)
	Middle (4–9 years) (Beginner (<3 years))	−9.716 (0.030)**	−4.480 (0.263)	−0.318 (0.678)	−1.792 (0.535)	4.145 (0.098)*	−3.818 (0.241)
Driving frequency	Driving every week (Rarely driving)	−0.852 (0.840)	8.108 (0.033)**	−0.980 (0.184)	0.100 (0.967)	−0.826 (0.730)	2.041 (0.515)
Warning display types * Weather condition	HUD*Foggy weather (HUD*Clear weather)	−7.724 (0.032)*	−7.800 (0.035)**	−0.183 (0.768)	1.385 (0.646)	0.626 (0.784)	0.153 (0.966)
	HDD*Foggy weather (HDD*Clear weather)	−4.170 (0.371)	−9.181 (0.011)**	−0.184 (0.799)	6.171 (0.038)**	1.207 (0.570)	−0.529 (0.854)
	Baseline*Foggy weather	0^a	0^a	0^a	0^a	0^a	0^a
	Baseline*Clear weather	0^a	0^a	0^a	0^a	0^a	0^a
Warning display types * Gender	HUD*Female (HUD*Men)	−3.466 (0.431)	0.325 (0.942)	−0.503 (0.503)	−8.761 (0.019)**	2.873 (0.307)	−9.125 (0.038)**
	Baseline*Female	0^a	0^a	0^a	0^a	0^a	0^a
	Baseline*Male	0^a	0^a	0^a	0^a	0^a	0^a
Warning display types * Driving experience	HUD*Middle (4–9 years) (HUD*Beginner (<3 years))	7.359 (0.076)*	5.169 (0.226)	0.011 (0.988)	0.966 (0.782)	−1.453 (0.583)	4.659 (0.262)
	HDD*Middle (4–9 years) (HDD* Beginner (<3 years))	12.711 (0.019)**	8.513 (0.042)**	−1.449 (0.085)*	0.727 (0.830)	−6.036 (0.014)**	5.899 (0.080)*
	Baseline*Senior (>10 years)	0^a	0^a	0^a	0^a	0^a	0^a
	Baseline*Middle (4–9 years)	0^a	0^a	0^a	0^a	0^a	0^a
	Baseline*Beginner (<3 years)	0^a	0^a	0^a	0^a	0^a	0^a
Weather conditions * Gender	Foggy weather*Female (Foggy weather*Male)	1.209 (0.684)	1.422 (0.701)	1.206 (0.026)**	0.629 (0.819)	−0.590 (0.782)	1.901 (0.546)
	Clear weather*Female	0^a	0^a	0^a	0^a	0^a	0^a
	Clear weather*Male	0^a	0^a	0^a	0^a	0^a	0^a
Gender * Driving experience	Female*Senior (>10years) (Female*Beginner (<3years))	−6.085 (0.192)	−20.252 (<0.001)**	0.146 (0.869)	2.562 (0.539)	7.843 (0.032)**	−7.941 (0.112)
	Female*Middle (4–9 years) (Female*Beginner (<3years))	1.221 (0.750)	−3.187 (0.506)	−1.173 (0.082)*	−0.796 (0.817)	3.293 (0.221)	−7.308 (0.069)*
	Male*Senior (>10 years)	0^a	0^a	0^a	0^a	0^a	0^a
	Male*Middle (4–9 years)	0^a	0^a	0^a	0^a	0^a	0^a
	Male*Beginner (<3 years)	0^a	0^a	0^a	0^a	0^a	0^a
Gender * Driving frequency	Female*Driving every week (Female*Rarely driving)	3.484 (0.540)	14.468 (0.051)*	−1.198 (0.274)	−5.080 (0.376)	−9.071 (0.037)**	4.018 (0.509)
	Male*Driving every week	0^a	0^a	0^a	0^a	0^a	0^a
	Male*Rarely driving	0^a	0^a	0^a	0^a	0^a	0^a
Driving frequency* Driving experience	Driving every week*Senior (>10 years) (Driving every week*Beginner (<3 years))	−0.198 (0.958)	−14.146 (0.004)**	2.321 (0.002)**	5.790 (0.090)*	0.981 (0.716)	−1.222 (0.761)
	Rarely driving*Senior (>10 years)	0^a	0^a	0^a	0^a	0^a	0^a
	Rarely driving*Middle (4–9 years)	0^a	0^a	0^a	0^a	0^a	0^a
	Rarely driving*Beginner (<3 years)	0^a	0^a	0^a	0^a	0^a	0^a

*Significant at 10% (bold values).
**Significant at the 5% (bold values).
0^aReference item.

4.2.1.2.1. Maximum deceleration. Taking the Baseline group as the reference, the absolute value of the maximum deceleration of the HUD and HDD groups decreased significantly (HUD: $p < 0.001$; HDD: $p = 0.018$); of these, the HUD group decreased the most. The absolute value of the maximum deceleration of drivers with senior driving experience was greater than drivers with beginner driving experience ($p = 0.012$). In the interaction between weather conditions and gender, taking clear weather as the reference, the absolute value of the maximum deceleration taken by females was smaller than that of males in foggy weather ($p = 0.026$). In the interaction between driving frequency and driving experience, taking drivers who rarely drive as the reference, the absolute value of the maximum deceleration of drivers with senior driving experience was smaller than drivers with beginner driving experience among drivers who drive every week ($p = 0.002$).

4.2.1.2.2. Braking distance. Taking the Baseline group as the reference, the braking distances of the HUD and HDD groups were longer (HUD group: $p < 0.001$; HDD group: $p = 0.008$); of these, the HUD group's were the longest. The braking distance of drivers with senior driving experience was shorter than drivers with beginner driving experience ($p = 0.030$). The farther position of accelerator-pedal release and first braking during the perception and decision stages make it necessary to take more urgent braking actions during the risk-avoidance manipulation phase. Taking the Baseline group as the reference, the interaction between warning display types and weather conditions showed that the braking distance in foggy weather was longer than that in clear weather in the HDD group ($p = 0.038$); and the interaction between warning display types and gender, showed that the braking distance of female drivers was shorter than that of male drivers in the HUD group ($p = 0.019$).

4.2.1.3. Risk-avoidance result stage

Greater minimum collision distances and an earlier position of the minimum speed indicated better risk-avoidance results (Z. Yang et al., 2019).

4.2.1.3.1. Minimum collision distance. Taking the Baseline group as the reference, the minimum conflict distances of the HUD and HDD groups were longer (HUD: $p < 0.001$; HDD: $p = 0.007$); of these, the HUD group's were the longest. In the interaction between warning display types and driving experience, taking the Baseline group as the reference, the minimum conflict distance of drivers with middle driving experience was smaller than that of drivers with beginner driving experience in the HDD group ($p = 0.014$). Taking male as the reference, the interaction between gender and driving experience showed that females with senior driving experience have a greater minimum conflict distance than drivers with beginner driving experience ($p = 0.032$); the interaction between gender and driving frequency showed that females who drive every week have a lower minimum conflict distance than drivers who rarely drive ($p = 0.037$).

4.2.1.3.2. Position of minimum speed. Taking the Baseline group as the reference, the positions of minimum speed of the HUD and HDD groups were earlier (HUD: $p = 0.018$; HDD: $p = 0.042$); of these, the HUD group's were the earliest. Taking the Baseline group as the reference, the interaction between warning display types and gender showed that females reached the position of minimum speed earlier than males in the HUD group ($p = 0.038$).

4.2.2. Analysis of influencing factors

Analysis of influencing factors refers to analyzing the overall variation patterns of the three influencing factors of warning display type, weather, and driver attributes in three stages in Table 8.

4.2.2.1. Warning display types. It can be seen that HUD and HDD in the connected environment help drivers allocate the distance spent in the three stages more reasonably, helping them choose better braking risk-avoidance strategies, and that HUD performs better than HDD. Specifically, HUD can shorten perceived reaction distance and greatly enhance the driver's ability to perceive danger. It enables the driver to appropriately extend the braking distance and adopt a more comfortable deceleration behavior under current conditions. At the same time, it can increase the minimum conflict distance and effectively improve the ultimate level of safety. There is evidence that HUD pedestrian warnings change the braking process from a panic brake (slow reaction, hard brake) to a comfortable brake (quick reaction, gentle brake).

4.2.2. 2. Weather conditions. Weather conditions are also an important factor affecting the braking risk-avoidance strategy. Research shows that fog absents a separate main effect on the six indicators of braking risk-avoidance strategy. There are two explanations for this phenomenon: First, it may be related to the level of fog. This paper only considered dense fog with visibility of 240 m. Whether there are significant differences under fog conditions with lower visibility, needs to be further explored. Second, it may be that drivers are more attentive and careful in foggy weather (Wu et al., 2018; Wu et al., 2018). Drivers' more cautious behavior in fog may offset the reduction in driving performance caused by the reduced visibility. Although there is no significant main effect, weather conditions and warning display types showed a significant interaction effect. HDD and HUD warnings improved the drivers' braking risk-avoidance strategy under fog conditions, especially helping the driver to complete the perception decision stage earlier, leaving more space for the driver to adjust the manipulation behavior. However, the failure to improve the final avoidance outcome may be due to the milder braking process offsetting the advantages of the risk-avoidance result stage.

4.2.2.3. Driver attributes. Among the driver attributes, the main effect of gender on the six indicators of braking risk-avoidance strategy was not significant (Wu et al., 2018).

Compared to drivers with beginner driving experience who rarely drive, drivers with senior driving experience who drive every week tend to engage in braking behaviors later and more urgently. It is worth noting that compared to drivers with beginner driving experience, drivers with middle driving experience choose to take braking action earlier and complete danger avoidance earlier. The warning display types have different effects on drivers with different attributes. In this regard, HUD effectively improved females' final risk-avoidance ability; HUD and HDD increased the driving confidence of drivers with middle driving experience and enabled them to choose more confident braking avoidance strategies (Luzuriaga et al., 2022).

To sum up, at the application level, the application of HUD warning in emergencies is worth promoting, especially in high-risk scenarios where the driver's line of sight is obstructed, such as hidden pedestrian crossing event outside the zebra crossing. When the driver's perception and decision abilities decrease due to external conditions (such as foggy weather), the setting of HDD and HUD warnings will be more valuable. In the promotion process of HUD, it is recommended to provide pre use training for drivers to increase their trust and familiarity with HUD warnings. At the same time, different warning forms and timing can be set for drivers with different attributes to meet the personalized needs of each driver. Before HUD can be widely promoted, it is necessary to conduct sufficient effectiveness testing on HUD. Additionally, during the testing process, the impact of HUD warning on driver perception and decision, risk-avoidance manipulation, and risk-avoidance result stages should be considered. At least 1–2 appropriate indicators should be selected for each stage, such as position of accelerator pedal release and position of first braking (perception and decision stage), braking distance and maximum deceleration (risk-avoidance manipulation stage), minimum conflict distance or position of minimum speed (risk-avoidance result stage), etc.

4.3. Limitations and future research

This paper also has the following limitations and related work to be done in the future. First, with the development of augmented reality (AR), as a new visual display type, AR-HUD has recently been used to release navigation information or warning information, and it has been considered that it may play a greater role in the future (Frémont et al., 2020; Hwang et al., 2015; Schwarz & Fastenmeier, 2017). This paper only studies HDD and HUD, and the influence of AR-HUD will be studied in the future. Second, fog was often used in previous studies as a typical case of reduced visibility. This paper has explored the impact of dense fog on driver braking risk-avoidance strategies. In the future, research on different levels of foggy weather and other adverse weather conditions (e.g., rain and snow) can be added. The third is the difference between the experimental results of the driving simulator and the real-vehicle test. This difference is inevitable, but it can be reduced by calibration. In the future, we can calibrate this difference through real-vehicle testing, and the experimental results after calibration will be more realistic. Finally, the learning effect is also a common issue in driving simulation experiments. Although the learning effect of this paper is not significant and will not affect the experimental results through testing, it still exists. In future experiments, we can try to avoid learning effects as much as possible by increasing the complexity of the scenario and the rest time of the driver. And the learning effect will be measured through subjective questionnaires as an auxiliary measure.

5. Conclusion

In the high-risk scenario of hidden pedestrian crossing event outside the zebra crossing, this paper studied the influence of three warning display types (Baseline/HDD/HUD) combined with two weather conditions (clear weather/foggy weather) on a driver's braking risk-avoidance strategy. It also considered the driver's attributes as factors. The driving behavior data of thirty-four participants were obtained through a driving simulator. Based on the description of the process, an index system of braking risk-avoidance strategies was established. A GLMM was used to explore the influence of different factors (warning display types, weather conditions and driver attributes) on the six indicators, the differences between different levels of the same factor, and the interaction effects between different factors.

The results showed that warning display types, driving experience and driving frequency had a significant main effect on the braking risk-avoidance strategy. Compared to the Baseline conditions and HDD warning, HUD warning helps drivers choose the optimal risk avoidance strategy. Specifically, the drivers reacted at positions closer to the starting point; braked more smoothly and gently; and maintained a greater safety margin between themselves and the pedestrian. Drivers with senior driving experience and who drive every week tended to spend more time in the perception and decision stage to observe the changes in the hazard.

Warning display type and weather conditions, driving experience and driving frequency showed interaction effects on braking and risk-avoidance strategies. The HUD and HDD in foggy weather significantly improved the driver's perception and decision ability. Females braked more urgently and completed risk-avoidance earlier when using HUD. Drivers with middle driving experience tended to adopt confident strategies when using HDD. Warning display types may be particularly useful for inclement weather and different types of drivers.

The contribution of this paper can be summarized as the following three points: First of all, the HUD test platform in the connected environment constructed by the driving simulator can be widely used to test the HUD effectiveness in other safety-critical events (e.g., left-side vehicle merging and conflict with turning left), and it can also be used to carry out AR-HUD related tests in the future. Secondly, based on the detailed description of the braking risk-avoidance process under pedestrian crossing events outside the zebra crossing, an index system for studying the braking risk-avoidance strategy was constructed. Ultimately, a general research method for testing and evaluating the effectiveness of the HUD warning system in such braking risk-avoidance events was formed. The research results support the development and implementation of design regulations for HUD, provide a reference for the rationalization and personalized design of HUD warning systems.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This study is supported by the Beijing Natural Science Foundation Fengtai Rail Transit Frontier Research Joint Foundation (Grant No. ZD038001202201).

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